

1. Algebra and functions

What students need to learn:

Laws of indices for all rational exponents.	The equivalence of $a^{m/n}$ and $\sqrt[n]{a^m}$ should be known.
Use and manipulation of surds.	Students should be able to rationalise denominators.
Quadratic functions and their graphs.	
The discriminant of a quadratic function.	
Completing the square. Solution of quadratic equations.	Solution of quadratic equations by factorisation, use of the formula and completing the square.
Simultaneous equations: analytical solution by substitution.	For example, where one equation is linear and one equation is quadratic.
Solution of linear and quadratic inequalities.	For example, $ax + b > cx + d$, $px^2 + qx + r \geq 0$, $px^2 + qx + r < ax + b$.
Algebraic manipulation of polynomials, including expanding brackets and collecting like terms, factorisation.	Students should be able to use brackets. Factorisation of polynomials of degree n , $n \leq 3$, e.g. $x^3 + 4x^2 + 3x$. The notation $f(x)$ may be used.
Simple algebraic division; use of the Factor Theorem and the Remainder Theorem.	Only division by $(x + a)$ or $(x - a)$ will be required. Students should know that if $f(x) = 0$ when $x = a$, then $(x - a)$ is a factor of $f(x)$. Students may be required to factorise cubic expressions such as $x^3 + 3x^2 - 4$ and $6x^3 + 11x^2 - x - 6$. Students should be familiar with the terms 'quotient' and 'remainder' and be able to determine the remainder when the polynomial $f(x)$ is divided by $(ax + b)$.

Graphs of functions; sketching curves defined by simple equations.

Geometrical interpretation of algebraic solution of equations. Use of intersection points of graphs of functions to solve equations.

Functions to include simple cubic functions and the reciprocal function

$$y = \frac{k}{x} \text{ with } x \neq 0.$$

Knowledge of the term asymptote is expected.

Also $y = a^x$ and its graph and trigonometric graphs – see sections 4 and 5.

Knowledge of the effect of simple transformations on the graph of $y = f(x)$ as represented by $y = af(x)$, $y = f(x) + a$, $y = f(x + a)$, $y = f(ax)$.

Students should be able to apply one of these transformations to any of the above functions (quadratics, cubics, reciprocal, sine, cosine, tangent and power functions of the type $y = a^x$) and sketch the resulting graph.

Given the graph of any function $y = f(x)$ students should be able to sketch the graph resulting from one of these transformations.
